FYS2160: Lab 2 - Rapport

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INTRODUCTION

Understanding the phase equilibrium of fluids is a fundamental aspect of thermodynamics and has various practical applications in a wide variety of fields. One of the essential thermodynamic properties related to phase transitions is the latent heat, which characterizes the energy exchange during the transition between two phases, such as between a liquid and a gas. By looking at the liquid-gas phase equilibrium of a van der Waals fluid, and comparing its heat of vaporization to that of water, it is then possible to evaluate its accordance with Trouton's rule - a rule suggesting that the entropy of vaporization is $\approx 11R$, where R is the gas constant.

Water is of course a unique and vital substance with an exceptionally high latent heat of vaporization compared to most other liquids. Its molar latent heat of vaporization far exceeds the prediction given by Trouton's rule. This distinctive property is primarily due to the arrangement of hydrogen bonds and the formation of a highly ordered and structured network of molecules, which contribute to its unique properties - most notably its high latent heat of vaporization.

In this study, the aim is to explore the liquid-gas phase equilibrium of the van der Waals fluid both analytically and numerically, comparing its latent heat of vaporization to that of water. Additionally, molecular dynamics (MD) simulations will be employed to investigate the behavior of ice, liquid water, and water vapor at the microscopic level. Through these comprehensive analyses, the focus will be to understand the relationship between order, entropy, and latent heat in the context of our experiments and Trouton's rule, providing valuable insights into the phase behavior of these substances.

METHODS

I. NUMERICAL PART

A. Molecular dynamics of ice, water and vapour

In this part we perform molecular dynamics simulations of the three phases of water.
Using LAMMPS we simulate the molecular structure of ice, water and gas, and we compare the movement, structure, energy and enthalpy of the different phases with each other and compare this to theoretical approximations.

B. van der Waals equation of state

1. *Analytical work*

Here we will derive analytical expressions for the van der Waals equation of states, and see if van der Waals fluid liquid – gas phase transition follows the Trouton rule.

The Helmholtz free energy for the van der Waals fluid is

$$F = -NkT\left(\ln\left(\frac{n_q(V-Nb)}{N}\right)+1\right) - \frac{aN^2}{V} \tag{1}$$

where N is the number of molecules, k is the Boltzmann constant, T is temperature, $n_Q(T) = (2\pi mkT/h^2)^{3/2}$ is a function of T only, V is volume, b is the excluded volume per particle and a is a constant related to the average binding energy.

The pressure is found as

$$\begin{aligned} P &= -\left(\frac{\partial F}{\partial V}\right)_{T,N} \\ &= NkT\frac{\partial}{\partial V}\left(\ln\left(\frac{n_q(V-Nb)}{N}\right)\right) + aN^2\frac{\partial}{\partial V}\left(\frac{1}{V}\right) \\ &= NkT\left(\frac{N}{n_q(V-Nb)}\cdot\frac{n_q}{N}\right) + aN^2\left(-\frac{1}{V^2}\right) \\ &= \frac{NkT}{V-Nb} - \frac{aN^2}{V^2}. \end{aligned}$$

The Gibbs free energy can thus be written as

$$\begin{aligned} G &= F + PV \\ &= -NkT\left(\ln\left(\frac{n_q(V-Nb)}{N}\right)+1\right) - \frac{aN^2}{V} + PV. \end{aligned}$$

By using the reduced parameters

$$\hat{V} = V/V_c, \quad \hat{T} = T/T_c, \quad \hat{P} = P/P_c,$$

where the critical parameters are

$$P_c = \frac{a}{27b^2}, \quad V_c = 3Nb, \quad kT_c = \frac{8a}{27b},$$

we can rewrite Gibbs free energy as

$$G = -NkT\left(\ln\left(\frac{n_q Nb(3\hat{V}-1)}{N}\right)+1\right) - \frac{aN^2}{3bN\hat{V}} + \hat{P}\hat{V}\frac{3abN}{27b^2}$$

$$\frac{G}{NkT_c} = -\frac{NkT}{NkT_c}\left(\ln\Big(n_q b(3\hat{V}-1)\Big)+1\right) - \frac{aN}{3b\hat{V}NkT_c} + \hat{P}\hat{V}\frac{Na}{9bNkT_c}$$

$$\frac{G}{NkT_c} = -\frac{NkT}{NkT_c}\left(\ln\Big(3\hat{V}-1\Big)+f(T)+1\right) - \frac{a}{3b\hat{V}kT_c} + \hat{P\hat{V}}\frac{a}{9bkT_c}$$

$$\frac{G}{NkT_c} = -\hat{T}\ln\Big(3\hat{V}-1\Big) - \frac{27ab}{24ab\hat{V}} + \hat{P}\hat{V}\frac{27ab}{72ab} + f(\hat{T})$$

$$\frac{G}{NkT_c} = -\hat{T}\ln\Big(3\hat{V}-1\Big) - \frac{9}{8\hat{V}} + \frac{3}{8}\hat{P}\hat{V} + f(\hat{T})$$

$$\hat{G} = \frac{8G}{3NkT_c} = -\frac{8}{3}\hat{T}\ln\Big(3\hat{V}-1\Big) - \frac{3}{\hat{V}} + \hat{P}\hat{V} + f(\hat{T}).$$

In similar fashion we can rewrite the pressure as

$$\hat{P} = \frac{P}{P_c}$$

$$= \left(\frac{N\hat{T}\frac{8a}{27b}}{Nb(3\hat{V}-1)} - \frac{aN^2}{(3Nb\hat{V})^2}\right)\frac{27b^2}{a}$$

$$= \left(\frac{8\hat{T}Na}{27Nb^2(3\hat{V}-1)} - \frac{aN^2}{9N^2b^2\hat{V}^2}\right)\frac{27b^2}{a}$$

$$= \frac{8\hat{T}}{3\hat{V}-1} - \frac{3}{\hat{V}^2}$$

2. Numerical work

In this section we want to find the coexistence line $\hat{P}_b(\hat{T}_b)$ by using a numerical estimation of the latent heat of a Van der Waals system from a fluid to a gas, using the relations derived in the analytical section.

By calculating and plotting $\hat{P}(\hat{V})$, $\hat{G}(\hat{V})$ and $\hat{G}(\hat{P})$ we determine the boiling temperature $\hat{T}_b$ by eye, and we use this to find the coexistence line.

From this we use the Clausius-Clapeyron relation

$$\frac{dP}{dT} = \frac{\Delta S}{\Delta V} = \frac{\Delta L}{T\Delta V}$$

and with the fact that at the boiling temperature the Gibbs free energy is zero $\Delta G = \Delta H - T\Delta S = 0$, and thus the latent heat L is equal to the enthalpy of vaporization, and we get

$$H_v = \frac{dP}{dT}T\Delta V. \tag{2}$$

To determine $\frac{dP}{dT}$ we use a finite difference method, we estimate ΔV bye looking at out plot for $\hat{P}(\hat{V})$ at the pressure of the boiling-temperature.

II. EXPERIMENTAL PART: MELTING AND VAPORIZATION

1. Heat of vaporization

In this experiment, we aim to study the relationship between pressure and temperature. Some simplifications will be made : the molar volume of water will be ignored, water vapor will be considered as an ideal gas, and the latent heat will be assumed to be constant over small time increments.

We will initiate the experiment at a pressure of 57 KPa, incriminating the pressure by 5 KPa in each step until atmospheric pressure of approximately 100 KPa is reached. The pressure will be increased by opening the valve labeled H2, allowing air to slowly enter the chamber.

At each step, we will wait for the water to begin boiling before recording the data.

To find the enthalpy of vaporisation we use the Clausius-Clapeyron relation

$$\ln\frac{P_1}{P_2} = -\frac{H_v}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \tag{3}$$

where R is the gas constant.
By using linear regression on the logarithmic relationship between the measured pressures P and temperatures T, we find the value $-H_v/R$ as the slope of the linear model.

2. Heat of melting

Now we want to determine the latent heat of *melting* using a calorimeter of stainless steal with a plastic lid.

We start by filling the calorimeterer with 1.25 l of water at room temperature. From here we add a current I of 0.75 A and a voltage U of 90 V and record the temperature increase until the water reach $T_1 = 29.18°$ C. From this we calculate the heat capacity as

$$C_0 = \frac{UI}{dT/dt}. \tag{4}$$

Now we weigh an ice cube of $m = 250$ g at temperature $T_0 = 0°$ C before putting it directly in the calorimeter, where we measure the temperature change until it stabilize at $T_2 = 13.04°$ C after around 5 minutes. From this we calculate the enthalpy of melting as

$$H_m = \frac{C_0}{m}(T_1 - T_2) - C_v(T_2 - T_0) \tag{5}$$

RESULTS AND DISCUSSION

NUMERICAL PART

A. Molecular dynamics of ice, water and vapor

The LAMMPS simulation of the ice-molecules is visualized in FIG 1 from Ovito. Here we clearly see a structured pattern, which is consistently held throughout the simulation. This is compatible with what we would expect from a solid. In FIG 2 we see a more unstructured

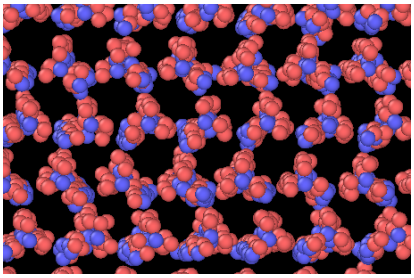


FIG. 1. The Ovito visualization of the molecular structure in the ice phase. We see a very structured molecule placement which we would expect from a solid.

placement of the molecules, which is what we would expect from a fluid. In FIG 3 we see how the gas phase

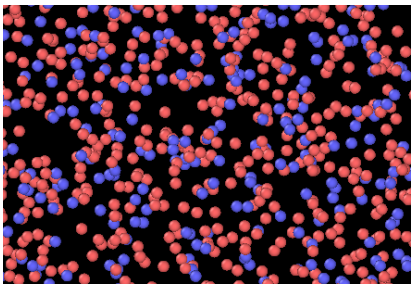


FIG. 2. The Ovito visualization of the molecular structure in the fluid phase. The molecular placement is a lot less structured compared to the ice-phase.

causes a very unstructured molecule placement, where the molecules are far away from each other.

Upon plotting the `log`-file for ice/water at $300\,K$, fetching relevant values and plotting the energies, the following plot (4) is found:

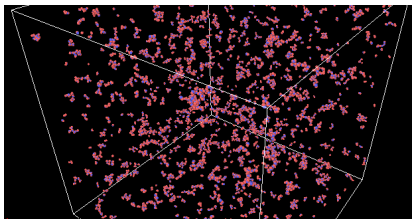


FIG. 3. The Ovito visualization of the molecular structure in the vapor phase. The structure is chaotic and the molecules are very far apart compared to the solid and fluid phase.

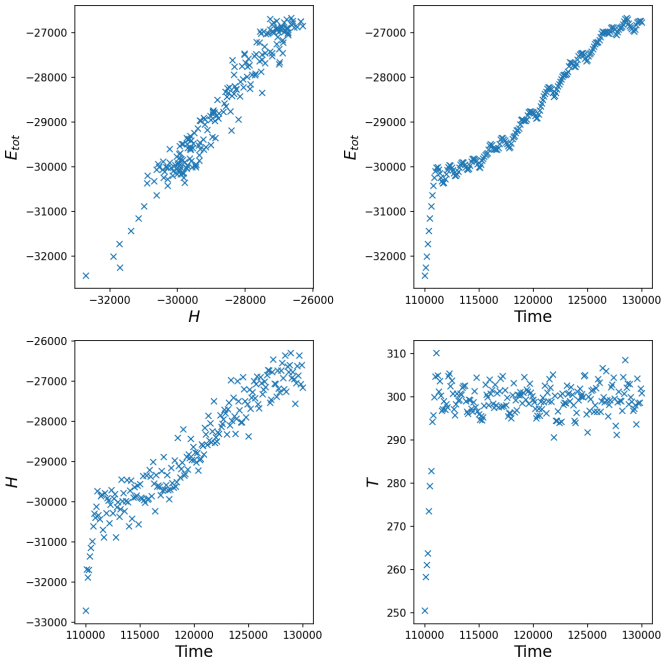


FIG. 4. Plot of the energies, and temperature, over time - describing a system of an iceblock melting into water, at room temperature.

From the plot a few things become apparent. In the upper-left plot, we see a rather linear distribution of our experimental data, which suggests that either the pressure or volume remain somewhat constant - which concurs. Since we know that depending on which ensemble you use, either the NPT-or NVT-ensemble, either the pressure or the volume will be constant. In the lower-right plot, we see a rise in the temperature in the that halts around $300\,K$ - here the ice block has reached thermal equilibrium with its surroundings, and there is no heat transfer occurring between them.

If we repeat the plot for the `log`-file for water at $400\,K$, the following plot (5) is found. Looking over the plots, we once more see a seemingly linear relation between the total energy and the enthalpy - which suggests that either the pressure or volume remain constant, like the previous plot. Since we here used the

NVT-ensemble and not the NPT-ensemble, we know that the volume remains constant.

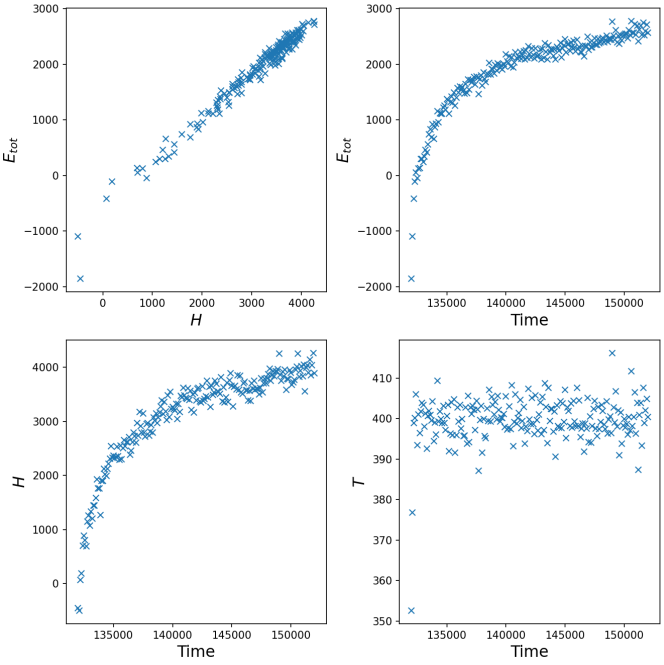


FIG. 5. Plot of the energies, and temperature, over time - describing a system of a water evaporating

B. van der Waals equation of state

After choosing values of $\hat{T}_b$, and a suitable range for our array of V-values, we plot the quantaties $\hat{P}(\hat{V})$, $\hat{G}(\hat{V})$ and $\hat{G}(\hat{P})$ where we find the coexistence pressure as

- For $t_b = 0.4$ self-intersection is when $P = 0.025$.
- For $t_b = 0.5$ self-intersection when $P = 0.05$
- For $t_b = 0.6$ self-intersection is when $P = 0.1$
- For $t_b = 0.7$ self-intersection is when $P = 0.2$
- For $t_b = 0.8$ self-intersection is when $P = 0.38$
- For $t_b = 0.9$ self-intersection is when $P = 0.65$

From this we plot the coexistence line $\hat{P}(\hat{T}_b)$ shown in FIG 12, and we see that the line resembles the figure in the lab text. Calculating the entropy H_v from equation 2 we get the graph shown in FIG 13. From this we see that the graph is somewhat around the theoretical value of 11, but it seems that our determinations of ΔV and dP/dT have been inaccurate.

The principle of corresponding states suggests that all fluids, when compared at the same reduced temperature and pressure, have similar behavior.

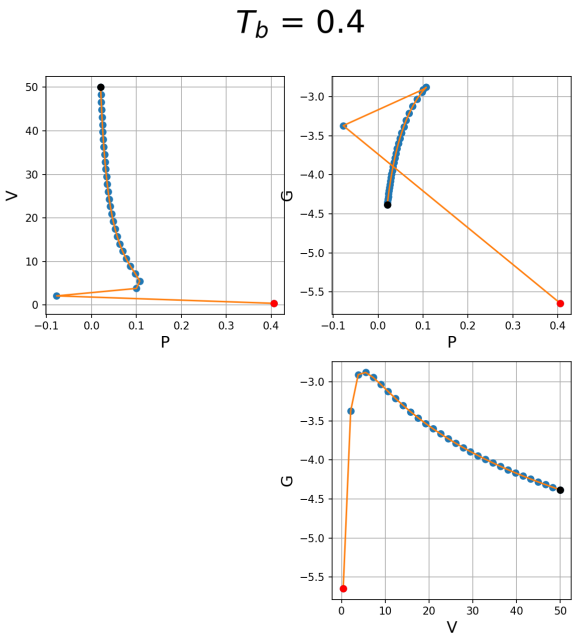


FIG. 6. $\Delta\hat{V} = 33$

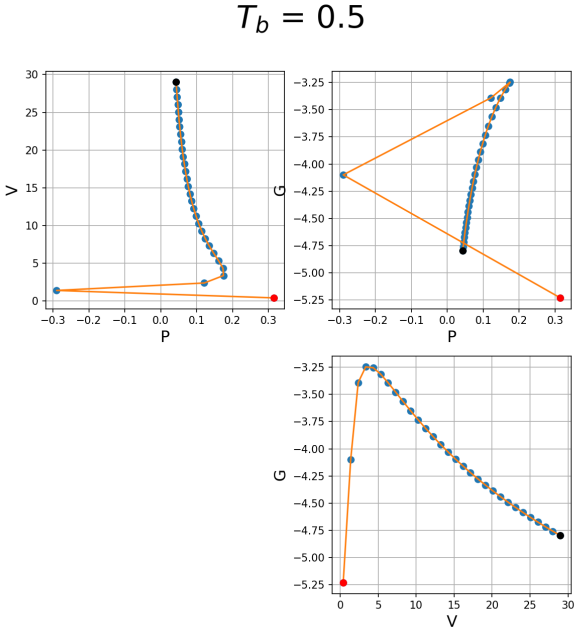


FIG. 7. $\Delta\hat{V} = 24$

For water, the critical temperature T_c is 647.096 K, and the critical pressure P_c is 22.064 MPa.

The reduced temperature (T_r) and reduced pressure (P_r) are defined as:

$$\hat{T} = \frac{T}{T_c}$$

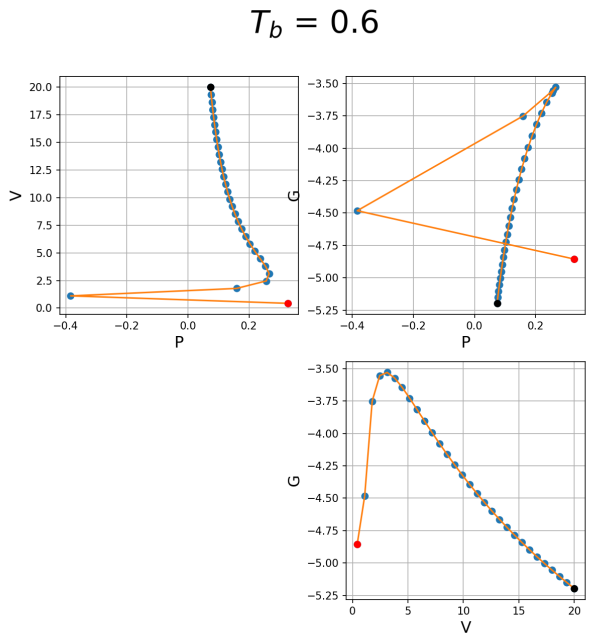


FIG. 8. $\Delta\hat{V} = 13.8$

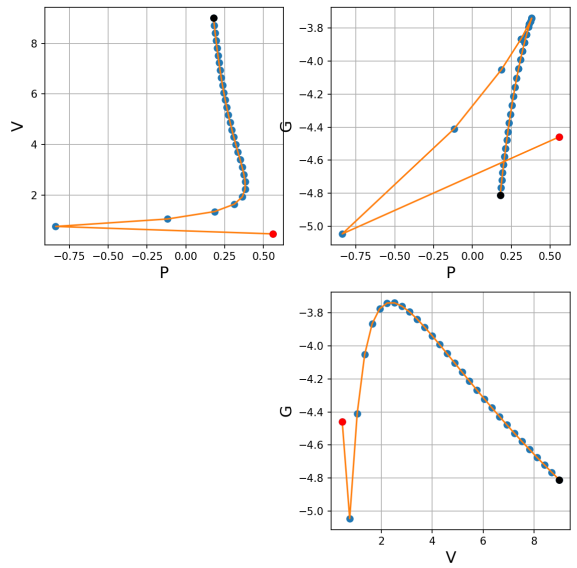


FIG. 9. $\Delta\hat{V} = 7.3$

$$\hat{P} = \frac{P}{P_c}$$

Normal pressure and temperature are :

- $P = 0.1MPa = 1bar \implies \hat{P} = 0.05$
- $T = 294k = 25C^\circ \implies \hat{T} = 0.45$

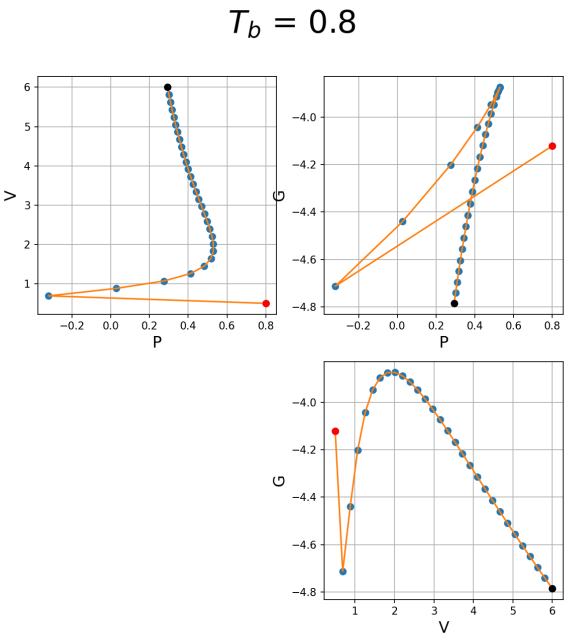


FIG. 10. $\Delta\hat{V} = 3.8$

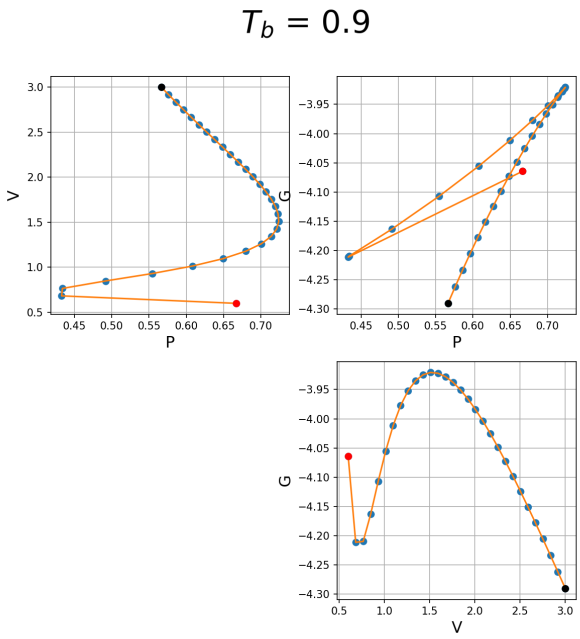


FIG. 11. $\Delta\hat{V} = 1.8$

It has been argued hat Trouton's rule follows from the principle of corresponding states. While the principle of corresponding states provides a sturdy framework to compare the behaviour of various substances, it does not *directly* explain Trouton's rule. Trouton's rule is an empirical observation, that states that the entropy of vaporisation for a wide range of substances is equal to a

step	T[C]	P[KPa]
1	82.8	57
2	85.1	62.6
3	87.3	67.8
4	89	73.2
5	91.2	78.6
6	92.9	84.4
7	94.7	89.7
8	97.5	95.3
9	98.6	100.6

TABLE I. The relationship between pressure and temperature

FIG. 12. The pressure coexistence line for different temperatures.

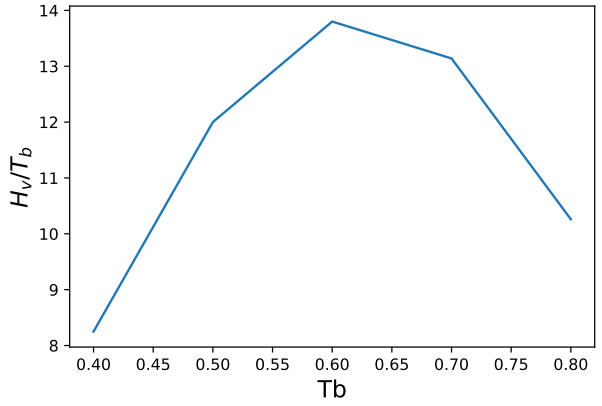


FIG. 13. The enthalpy divided by boiling temperature as a function of boiling temperature. The values fluctuate around the theoretical value of 11, but the determination of ΔV and the finite difference method for dP/dT seems to give inaccurate results.

Trouton's ratio, which is approximately 11 R. In other words, Trouton's rule is an empirical observation that applies to a specific property (heat of vaporization), and is not incumbent on the concept of corresponding states.

Another explanation for Trouton's rule using entropy change during vaporization can be expressed as the consistent increase in molecular freedom and energy that water gain when transitioning from the liquid to the vapor phase. This entropy change leads to the observation that $\frac{\Delta H}{T_b}$ is approximately constant for many substances.

EXPERIMENTAL

1. *Heat of vaporization*

By running "vannkoking.cap" program in the PC. We got the following readings showed in Table I: From the readings, the linear regression model yielded a slope of $\frac{H_v}{r} = -4990$. We obtained $r = 0.998$, which indicates the accuracy of the readings. In this case, the value is close to -1, indicating good accuracy. We calculated the enthalpy of vaporization to be 4980. We found the value of Trouton's rule as $\frac{H_v}{RT_b} = 13.13$. We observed a small deviation from Trouton's rule, which can be attributed to physical phenomena. The measurement will not be affected by the pressure of the water vapor above the water as long as the mixture achieves an equilibrium state. Adding cooling to the experiment will still gives good results, though it may extend the time needed to reach equilibrium. Measuring the temperature in the vapor phase provides a more accurate indication of the process properties. In the water phase, the results may be less accurate due to the inhomogeneous distribution of water temperature; the water will be heated more near the heat source. The vapor phase has a higher enthalpy than the water phase. Vaporization gives more degrees of freedom to the molecules, thereby increasing entropy, which in turn increases the enthalpy.

2. *Heat of melting*

By using the calorimeter to heat 1.25 l of water to a temperature $T_1 = 29.18°$ C, we measure a temperature increase of 0.0120 $\pm 2.0 \cdot 10^{-}6$ K/s. From 4 we calculate the heat capacity as

$$C_0 = \frac{90 \text{ V} \cdot 0.75 \text{ A}}{0.0120 \text{ K/s}} = 5625 \pm 1 \text{ J/K}.$$

With the spesific heat capacity of water $c_0 = 4.186$ J/gK, i.e. $C_0 = c_v m = 5233$ J/K our estimate is in the right order of magnitude, although not within the uncertainty.

With $T_2 = 13.04^\circ$ C we calculate the enthalpy of melting from equation 4 as

$$H_m = \frac{5233 \text{ J/K}}{250 \text{ g}}(29.18^\circ \text{ C} - 13.04^\circ \text{ C}) - 4200 \text{ J/kgK}(13.04^\circ \text{ C} - 0^\circ\text{C}) = 308.38 \text{ kJ}$$

As the theoretical value should be 333.55 kJ we see that we are in the right order of magnitude.

Compared to the enthalpy of melting, we see that the enthalpy of vaporization is much higher. This is because the phase shift from water to gas requires more energy to break the intermolecular forces than it does for the phase shift from ice to water.

CONCLUSION

In this study, we have delved into the fascinating realm of phase equilibrium in fluids, and investigated the importance of the role latent heat of vaporization plays in these transitions. The comparison between a van der Waals fluid and water, with its remarkable deviation from Trouton's rule, underscores the unique and exceptional nature of water due to its intricate arrangement of hydrogen bonds and molecular structure. Through a variety of analytical and numerical methods, along with molecular dynamics simulations, we have began to unravel the intricate relationship between entropy, and latent heat in these substances. Our study has provided valuable insights into the phase behavior of these fluids, shedding light on the fundamental principles of thermodynamics.